

SHIVANI SWAMINATHAN

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EDUCATION

Georgia Institute of Technology

Atlanta, GA

Bachelor of Science / Master of Science in Aerospace Engineering

B.S. Fall 2027 | M.S. Fall 2028

GPA: 3.8/4.0 | Dean's List | Relevant Coursework: Aerodynamics, Structural Analysis, Fluids, Thermodynamics, Mechanics of Materials

RESEARCH

Ben T. Zinn Combustion Laboratory

Atlanta, GA

Undergraduate Research Assistant

May 2026 – Present

- Improved **reacting-flow CFD model fidelity by 18%** across 8 steady and transient ANSYS Fluent cases through 19M-cell mesh refinement, combustion-model configuration, and high-fidelity multiphysics simulation.
- Collaborated with ANSYS engineers as technical partners** to validate 3 combustor cases using temperature, pressure, velocity, and species results, implementing 4 model and setup corrections to establish a reliable simulation baseline.

GT Aerospace Systems Design Laboratory (ASDL)

Atlanta, GA

Undergraduate Research Assistant – Systems Modeling

August 2025 – Present

- Developed an **executable SysML mobility subsystem** with requirements traceability across torque, traction, speed, power, and climb constraints; performed a 30+ case parametric trade study to establish feasible performance limits.
- Built a traceable SysML model integrating mission requirements, UAV system architecture, operational behavior, and performance equations; verified 3 delivery scenarios against range, battery, and payload requirements.

GT Aerospace Systems Design Laboratory (ASDL)

Atlanta, GA

Undergraduate Research Assistant – AI/ML

June 2026 – Present

- Translated lunar surface **CONOPS** into a Python AI agent that expanded a lunar-base simulation to 12+ autonomous construction tasks, automating landing-pad sequencing, resource allocation, and task dependencies for repeatable mission-plan evaluation.
- Integrated **4 lunar infrastructure models across 4 mission scenarios** by defining dependencies among landing-pad construction, in-situ resource utilization (ISRU), and power operations, enabling end-to-end systems simulation and mission analysis.

UGA Department of Physics and Astronomy

Athens, GA

Undergraduate Research Assistant

August 2024 – May 2025

- Analyzed **200+ stellar-flare light curves** in Python and modeled activity decay across 3 spectral classes in MATLAB, automating data processing to reduce turnaround time by 36% and earn an undergraduate research award.

EXPERIENCE

GT HyTech Racing

Atlanta, GA

Aerodynamics Engineer

August 2025 – Present

- Validated **carbon-fiber aerodynamic assemblies** across 6+ ride-height and angle-of-attack cases in ANSYS Fluent, quantifying up to 18% downforce and 11% drag variation to support competition design optimization.
- Reduced CFD analysis turnaround by 36%** by automating ANSYS Fluent post-processing in Python, enabling same-day comparison of 8+ aerodynamic designs and data-driven design decisions.

Handshake AI

Remote

CAD Specialist – OpenAI-Sponsored AI Evaluation Project

March 2026 – Present

- Developed **25+ CAD benchmark tasks** and weighted validation rubrics in SolidWorks, Siemens NX, and AutoCAD to evaluate AI-generated parts, assemblies, and engineering drawings against geometry, GD&T, and deliverable requirements.
- Evaluated 40+ CAD benchmarks and identified **60+ dimensional, manufacturability, and feature-quality issues**, improving requirements clarity, quality assurance, and scoring consistency before AI model testing.

PROJECTS

Wind Turbine Blade Fluid–Structure Interaction Analysis

July 2026

- Performed **multiphysics CFD/FEA analysis** across 6 operating conditions, linking aerodynamic loads to blade stress and deformation while verifying root radial force within 0.1% of analytical calculations.

Unified Aerodynamic Analysis Tool

August 2026

- Developed a **MATLAB aerodynamic analysis and optimization tool** combining Prandtl–Glauert-corrected thin-airfoil and shock-expansion theory; searched 45,000+ configurations across subsonic and supersonic regimes.

CubeSat Frame & Attitude Control Design

January 2025 – March 2025

- Reduced CubeSat frame mass by 10%** through ANSYS structural analysis against 5g stiffness, factor-of-safety, and buckling requirements; developed a MATLAB/Simulink PD controller achieving 3-axis stabilization.

SKILLS

Simulation: ANSYS Fluent (CFD), ANSYS Mechanical (FEA), Simscape Multibody

3D Modeling: SolidWorks (CSWA), CATIA, Inventor, AutoCAD, Fusion 360, Siemens NX

Systems Engineering: MagicDraw/Cameo (SysML), ModelCenter, Simulink

Programming: MATLAB (Certified MATLAB Associate), Python, Git